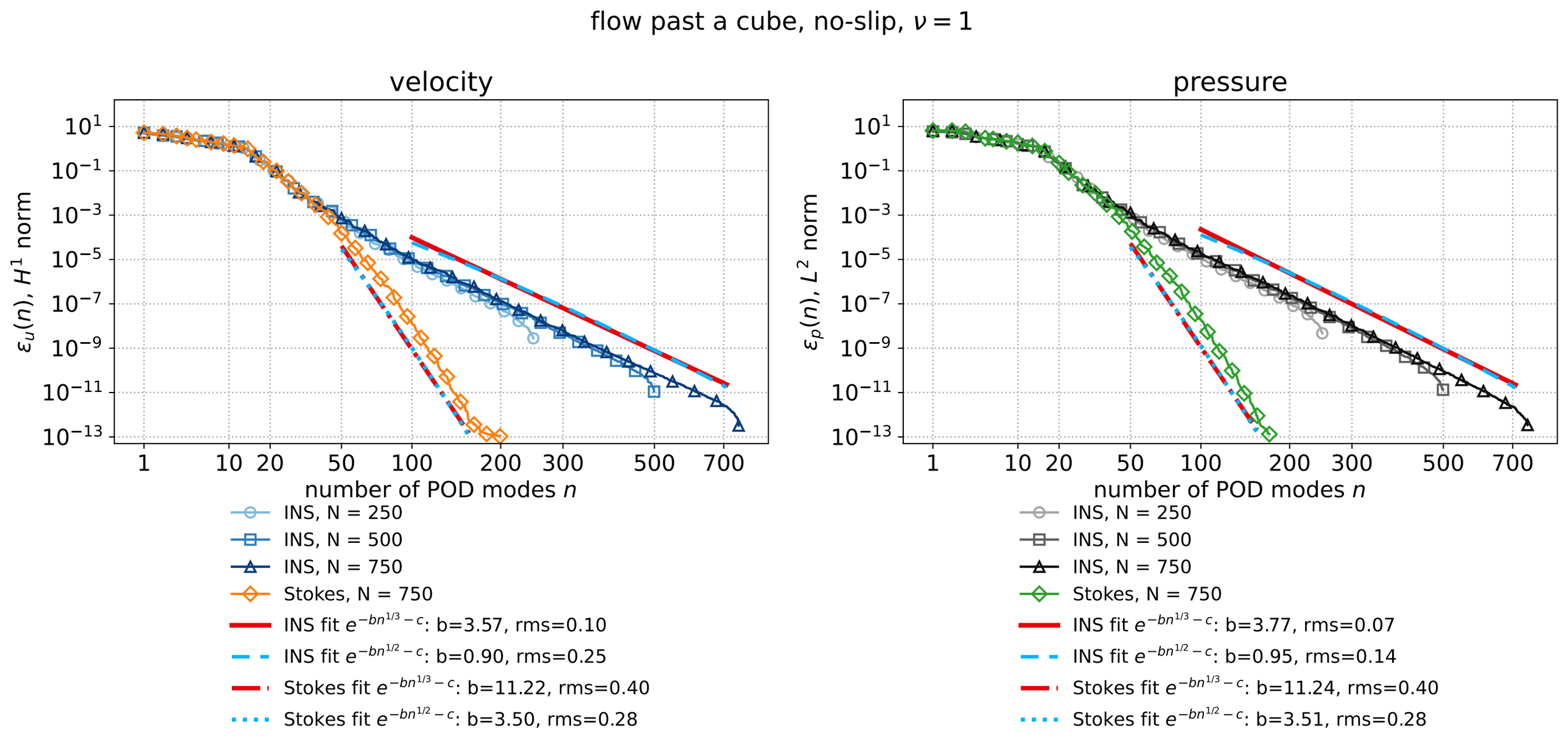

flow past a cube, no-slip, $\nu = 1$
velocity
pressure
$\varepsilon_u(n)$, H^1 norm
$\varepsilon_p(n)$, L^2 norm
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
1
10
20
50
100
200
300
500
700
number of POD modes n
INS, N = 250
INS, N = 500
INS, N = 750
Stokes, N = 750
INS fit $e^{-bn^{1/3}-c}$: b=3.57, rms=0.10
INS fit $e^{-bn^{1/2}-c}$: b=0.90, rms=0.25
Stokes fit $e^{-bn^{1/3}-c}$: b=11.22, rms=0.40
Stokes fit $e^{-bn^{1/2}-c}$: b=3.50, rms=0.28
INS, N = 250
INS, N = 500
INS, N = 750
Stokes, N = 750
INS fit $e^{-bn^{1/3}-c}$: b=3.77, rms=0.07
INS fit $e^{-bn^{1/2}-c}$: b=0.95, rms=0.14
Stokes fit $e^{-bn^{1/3}-c}$: b=11.24, rms=0.40
Stokes fit $e^{-bn^{1/2}-c}$: b=3.51, rms=0.28